

Dynamic Macroeconomics II

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Log-linearization and Uhlig's method

First Part

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Computational problems

- Macro research is about answering questions with dynamic, most of the time general equilibrium, models.
- These models cannot be solved by hand, except for very specific cases.
- Plus in macro for many questions we add uncertainty.
- In sum, (many times) we solve Dynamic Stochastic General Equilibrium Models (DSGE).

A categorization of problems and solution methods

	Deterministic	Stochastic
Linear	Blanchard-Kahn Uhlig's Undetermined Coefficients Uribe and Schmitt-Grohe	Same as previous column
Nonlinear	System of equations solver Shooting Dyn. Prog.	Dyn. Prog. (slow...) McGrattan's Weighted Residuals Study at MN, Penn, NYU...

Black-boxes versus flexibility

- A popular software to solve DSGE models is Dynare.
- **Benefit:** you give it the nonlinear system of equations, it linearizes for you, skipping algebra.
- Linearization: a first-order Taylor approximation of the nonlinear system.
- **Cost:** you do not know what is going on!
- Plus you need to know, for example, when doing a first-order, or second-order approximation matters (Dynare can quickly solve 1st and 2nd).
- My point of view: A pre-programmed, **rigid**, tool.
- For some uses it is perfectly OK.

What matters most

- The most important outcome of any solution method is the policy function: how endogenous variables are related to the **state variables**.
- Sometimes you can only get it numerically.
- If you have the policy function, you have everything.
- I.e you can calculate:
 - Impulse-response functions.
 - Statistics of model-predicted variables.
 - Transitions.
 - Simulate historical episodes.
 - Compute welfare.
 - Estimate parameter models (can check out McGrattan-Sargent; Dynare includes methods).

- We will learn to log-linearize the nonlinear system of equations that gives us the solution of a model.
- Then we will use Uhlig's Method of Undetermined Coefficients to solve the linear system.
- I will use an example from Uhlig to illustrate the steps.
- This example can be solved **by hand**.
- We will do that to understand how method works.
- Reference: Uhlig's chapter in Marimon and Scott (1999).

- We use the model in Uhlig (1999).
- It is a version of Hansen (1985).
- There is uncertainty about future exogenous productivity.
- Consumer solves:

$$\max E_0 \sum_{t=0}^{\infty} \beta^t [\log(C_t) - \lambda N_t]$$

s.t.

$$C_t + K_t = Y_t + (1 - \delta)K_{t-1}$$

$$Y_t = Z_t K_{t-1}^{\rho} N_t^{1-\rho}$$

$$\log(Z_t) = (1 - \psi) \log(\bar{Z}) + \psi \log(Z_{t-1}) + \varepsilon_t, \quad \varepsilon \sim (0, \sigma_{\varepsilon}^2)$$

Notation for capital:

- Uhlig: every period t , capital K_{t-1} is given, and capital K_t is chosen.
- We are used to take K_t as given, choose K_{t+1} .
- Keep difference in mind.

Plus one more thing:

- I am not writing uncertainty with precision.
- But you guys saw it in Dynamic Macro I.
- Cookbook:
 - You can write Lagrangian assuming certainty.
 - Take FOCs, obtain Euler equation(s).
 - Take expectation E_t of future variables.
 - Recall that $\text{cov}(X, Y) = E(XY) - E(X)E(Y)$.
 - I.e. $E(XY) \neq E(X)E(Y)$!

- Let Λ_t be the Lagrange multiplier associated to feasibility.
- Uhlig created variable R_t , not important.
- System of equations:

$$\frac{1}{C_t} = \Lambda_t$$

$$A = \Lambda_t(1 - \rho) \frac{Y_t}{N_t}$$

$$R_t = \rho \frac{Y_t}{K_{t-1}} + (1 - \delta)$$

$$Y_t = Z_t K_{t-1}^\rho N_t^{1-\rho}$$

$$C_t + K_t = Y_t + (1 - \delta)K_{t-1}$$

$$\Lambda_t = \beta E_t [\Lambda_{t+1} R_{t+1}]$$

$$\log(Z_t) = (1 - \psi) \log(\bar{Z}) + \psi \log(Z_{t-1}) + \varepsilon_t$$

- Linearize the model by using first-order Taylor approximation around point x_0 . Reminder:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

- For us point x_0 is steady state; plus we have functions of several variables.
- Write equations in the form $0 = F(X_t, Y_t)$.
- Let $\bar{X}$ be the steady-state value of variable X_t .
- Log-deviations from steady state:

$$x_t \equiv \log(X_t) - \log(\bar{X})$$

$$X_t = \bar{X}e^{x_t}$$

- Interpretation: multiply x_t by 100 gives us the % distance from steady state.

- Write F as a function of x_t, y_t : $F(x_t, y_t)$.
- We will linearize around steady state $x_t = y_t = 0$.
- Then in steady-state:

$$0 = F(0, 0) + F_x(0, 0)(x - 0) + F_y(0, 0)(y - 0).$$
- To simplify the algebra the following steady state expressions, in levels, are useful:

$$1 = \beta \bar{R}$$

$$\bar{R} = \rho \frac{\bar{Y}}{\bar{K}} + (1 - \delta)$$

- Steps:
 - 1 In (almost) every equation in system substitute $X_t = \bar{X}e^{x_t}$.
 - 2 Then linearize.
- Uhlig proposes an alternative to explicit differentiation: his “building blocks”, see his chapter, p. 34.

Log-linearizing

- Consider the first-order condition:

$$0 = \frac{1}{C_t} - \Lambda_t$$

$$0 = \frac{1}{\bar{C} e^{c_t}} - \bar{\Lambda} e^{\lambda_t} \quad \mathcal{F}(c_t, \lambda_t)$$

$$\mathcal{F}_{c_t}(0)! \quad \mathcal{F}_{\lambda_t}(0)!$$

Do first order Taylor:

$$0 = \frac{1}{\bar{C}} - \bar{\Lambda} + \frac{(-1)}{\bar{C}}(c_t - 0) - \bar{\Lambda}(\lambda_t - 0)$$

$$\frac{1}{\bar{C}} c_t = -\bar{\Lambda} \lambda_t \quad = 1/\bar{C} \text{ en ss}$$

$$\implies c_t = -\lambda_t$$

Handwritten notes: $\frac{1}{\bar{C}} = \bar{\Lambda}$ (from $\frac{1}{\bar{C}} - \bar{\Lambda} = 0$), $\frac{1}{\bar{C}} = \bar{\Lambda}$ (from $\frac{1}{\bar{C}} = \bar{\Lambda}$), $\frac{1}{\bar{C}} = \bar{\Lambda}$ (from $\frac{1}{\bar{C}} = \bar{\Lambda}$)

Obviously because we are approximating we should use $\approx$.

- Consumption-leisure equation:

$$0 = A - \Lambda_t(1 - \rho) \frac{Y_t}{N_t}$$

$$0 = A - \bar{\Lambda} e^{\lambda_t} (1 - \rho) \frac{\bar{Y} e^{y_t}}{\bar{N} e^{n_t}} \quad \mathcal{F}(\lambda_t, y_t, n_t)$$

Do first order Taylor:

$$0 = A - \bar{\Lambda}(1 - \rho) \frac{\bar{Y} \bar{\lambda}}{\bar{N}} - \bar{\Lambda}(1 - \rho) \frac{\bar{Y}}{\bar{N}} (\lambda_t - 0) - \bar{\Lambda}(1 - \rho) \frac{\bar{Y}}{\bar{N}} (y_t - 0) + \bar{\Lambda}(1 - \rho) \frac{\bar{Y}}{\bar{N}} (n_t - 0)$$

$\mathcal{F}_{\lambda_t}$ $\mathcal{F}_{y_t}$ $\mathcal{F}_{n_t}$

$$0 = \bar{\Lambda}(1 - \rho) \frac{\bar{Y}}{\bar{N}} (-\lambda_t - y_t + n_t)$$

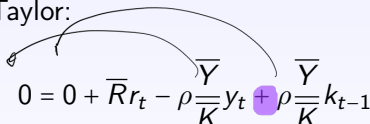
$$\implies n_t = \lambda_t + y_t$$

- Definition of payment to capital:

$$0 = R_t - \rho \frac{Y_t}{K_{t-1}} - (1 - \delta)$$

$$0 = \bar{R}e^{r_t} - \rho \frac{\bar{Y}e^{y_t}}{\bar{K}e^{k_{t-1}}} - (1 - \delta)$$

Do first order Taylor:



$$0 = 0 + \bar{R}r_t - \rho \frac{\bar{Y}}{\bar{K}} y_t + \rho \frac{\bar{Y}}{\bar{K}} k_{t-1}$$

$$\implies \bar{R}r_t = \rho \frac{\bar{Y}}{\bar{K}} (y_t - k_{t-1})$$

- Production function:

$$0 = \bar{Y}e^{y_t} - \bar{Z}e^{z_t}\bar{K}^\rho e^{\rho k_{t-1}}\bar{N}^{1-\rho}e^{(1-\rho)n_t}$$

Do first order Taylor:

$$0 = \bar{Y}y_t - \bar{Z}\bar{K}^\rho\bar{N}^{1-\rho}z_t - \bar{Z}\bar{K}^\rho\bar{N}^{1-\rho}\rho k_{t-1} - \bar{Z}\bar{K}^\rho\bar{N}^{1-\rho}(1-\rho)n_t$$

$$0 = \bar{Y}y_t - \bar{Z}\bar{K}^\rho\bar{N}^{1-\rho}(z_t + \rho k_{t-1} + (1-\rho)n_t)$$

$$\implies y_t = z_t + \rho k_{t-1} + (1-\rho)n_t$$

- Feasibility condition:

$$\begin{aligned}0 &= Y_t + (1 - \delta)K_{t-1} - C_t - K_t \\0 &= \bar{Y}e^{y_t} + (1 - \delta)\bar{K}e^{k_{t-1}} - \bar{C}e^{c_t} - \bar{K}e^{k_t}\end{aligned}$$

Do first order Taylor:

$$\begin{aligned}0 &= \bar{C}c_t + \bar{K}k_t - \bar{Y}y_t - (1 - \delta)\bar{K}k_{t-1} \\ \implies y_t &= \frac{\bar{C}}{\bar{Y}}c_t + \frac{\bar{K}}{\bar{Y}}(k_t - (1 - \delta)k_{t-1})\end{aligned}$$

- Euler equation:

$$\Lambda_t = \beta E_t(\Lambda_{t+1} R_{t+1})$$

$$0 = \bar{\Lambda} e^{\lambda_t} - \beta E_t(\bar{\Lambda} e^{\lambda_{t+1}} \bar{R} e^{r_{t+1}})$$

Do first order Taylor:

$$0 = \bar{\Lambda} \lambda_t - \beta \bar{R} E_t(\bar{\Lambda} \lambda_{t+1} + \bar{\Lambda} r_{t+1})$$

$$\implies \lambda_t = E_t(\lambda_{t+1} + r_{t+1})$$

- Finally, you do not need to linearize the AR(1) equation for Z_t . Simply rearrange terms:

$$Z_t = \psi Z_{t-1} + \varepsilon_t$$